**Parallel Processing – E-Commerce Sales Report**

# *HPC ISE-1*

***BTech. CSE Final Year***


| ***Name*** | ***PRN*** |
|---|---|
| *23510076* | *Arth Rai* |
| *23610067* | *Aditi Mittal* |
| *245200010* | *Siddheshwar Sakhare* |
| *245200009* | *Vedant Tathe* |


**Given:**

**An e-commerce company has 2,00,000 online orders. Each order contains:**

- **Order ID**
- **Customer ID**
- **Product ID**
- **Quantity**
- **Unit price**
- **Discount**
- **Delivery charge**

**The company wants to process all orders in parallel and generate a monthly sales report.**

**For every order, the system performs:**

1. **Calculate gross amount = quantity × unit price**
2. **Calculate discount amount**
3. **Calculate net order amount**
4. **Determine whether the order qualifies for free delivery**
5. **Calculate total sales for each product category**
6. **Find the best-selling product**
7. **Calculate the overall revenue**
8. **Generate the final sales report**

**Q1. Data Decomposition**

**Divide the 2,00,000 orders among 4 processors (P0–P3). Propose a suitable decomposition strategy.**

There are 2,00,000 orders and 4 processors. Therefore:

2,00,000 ÷ 4 = 50,000 orders per processor.

A suitable strategy is data decomposition, where each processor receives a different subset of orders.

| Processor | Orders Assigned |
|---|---|
| P0 | 1 – 50,000 |
| P1 | 50,001 – 1,00,000 |
| P2 | 1,00,001 – 1,50,000 |
| P3 | 1,50,001 – 2,00,000 |

**Q2. Classify the Data**

**Classify each item as Input, Intermediate, or Output:**

| Data Item | Classification | Reason |
|---|---|---|
| Quantity | Input | Given with each order. |
| Unit price | Input | Given with each order. |
| Gross amount | Intermediate | Calculated as quantity × unit price. |
| Discount amount | Intermediate | Calculated from discount information. |
| Net order amount | Intermediate | Calculated using gross amount and discount amount. |
| Free-delivery status | Intermediate | Determined from the order information/net amount. |
| Product-wise sales | Output | Final aggregated sales information. |
| Best-selling product | Output | Requires comparison of product sales. |
| Overall revenue | Output | Sum of all order revenues. |
| Final sales report | Output | Final generated report. |

**Q3. Processor-Level Processing**

**If P0 processes 50,000 orders, what intermediate results can P0 produce without communicating with other processors?**

If P0 processes 50,000 orders, it can independently calculate the following without communicating with P1, P2, or P3:

- Gross amount for each of its 50,000 orders.
- Discount amount for each order.
- Net amount for each order.
- Free-delivery status for each order.
- P0's total revenue.
- P0's product-wise sales totals.
- P0's locally best-selling product.
- Number of orders qualifying for free delivery.

Gross = Quantity × Unit Price

Net Amount = Gross Amount − Discount Amount

**Q4. Communication Analysis**

**Which of the following require combining results from P0, P1, P2 and P3?**

- **Net amount of an individual order**
- **Total revenue**
- **Best-selling product**
- **Free-delivery status**
- **Product-wise total sales**

**Give reasons.**

| **Operation** | **Combining P0–P3?** | **Reason** |
|---|---|---|
| Net amount of an individual order | No | The order is completely handled by the processor that owns it. |
| Total revenue | Yes | Each processor has only its local revenue; all four must be added. |
| Best-selling product | Yes | Each processor may have a different local best-selling product; global comparison is required. |
| Free-delivery status | No | Determined independently for each order. |
| Product-wise total sales | Yes | Sales of the same product may exist on multiple processors and must be combined. |

Total revenue, best-selling product, and product-wise total sales require aggregation across processors. Individual-order net amount and free-delivery status can be computed locally.

**Q5. Challenge Question**

**Suppose:**

- **P0 finds the highest revenue = ₹18 lakh**
- **P1 finds the highest revenue = ₹21 lakh**
- **P2 finds the highest revenue = ₹17 lakh**
- **P3 finds the highest revenue = ₹24 lakh**

**What additional operation is required to obtain the global highest revenue**

Each processor independently processes its assigned orders and calculates the maximum revenue within its own set of orders. These are called local maximum values:

P0 → ₹18 lakh

P1 → ₹21 lakh

P2 → ₹17 lakh

P3 → ₹24 lakh

However, these values represent only the maximum revenue found by each individual processor. They do not directly give the maximum revenue among all 2,00,000 orders.

Therefore, an additional global reduction operation using the MAX function is required. The local maximum values from P0, P1, P2, and P3 are communicated to a processor or a reduction stage, where they are compared:

Since ₹24 lakh is the largest value, the global maximum revenue is ₹24 lakh, and it was found by P3.

This process is called parallel reduction because each processor first performs a computation on its own portion of the data and produces a local result. These local results are then combined using an operation such as MAX, SUM, MIN, or COUNT to obtain the final global result.